

1. Determine if each of the following functions is a growth exponential or a decay exponential.

a. $y = 4^{-1.5x}$

b. $y = 3e^{-5x}$

c. $y = 2^{0.4x}$

d. $y = 0.6^{3x}$

a. The function $y = 4^{-1.5x}$ is a (1) _____ exponential.

b. The function $y = 3e^{-5x}$ is a (2) _____ exponential.

c. The function $y = 2^{0.4x}$ is a (3) _____ exponential.

d. The function $y = 0.6^{3x}$ is a (4) _____ exponential.

(1) ☐ decay
☐ growth

(2) ☐ growth
☐ decay

(3) ☐ growth
☐ decay

(4) ☐ growth
☐ decay

2. a. Graph the function $f(x) = e^x$ on the window $[-5, 5]$ by $[-10, 30]$.

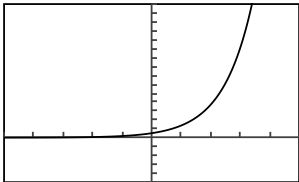
b. Find $f(1)$, $f(-5)$, and $f(3)$, rounded to three decimal places.

c. What is the horizontal asymptote of the graph?

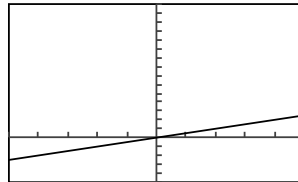
d. What is the y-intercept?

a. Choose the correct graph below.

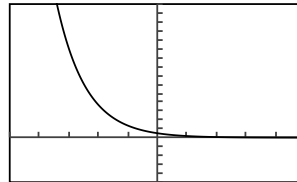
☐ A.



☐ B.



☐ C.



b. $f(1) =$ (Round to three decimal places as needed.)

$f(-5) =$ (Round to three decimal places as needed.)

$f(3) =$ (Round to three decimal places as needed.)

c. The horizontal asymptote of the graph is (1) _____

d. The y-intercept is .
(Type an ordered pair.)

- (1) ☐ x-axis.
☐ y-axis.
☐ $x = -2$.
☐ $x = 1$.

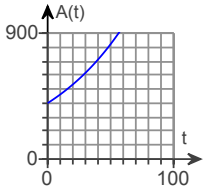
3. The amount of a radioactive isotope present at time t is given by $A(t) = 800e^{-0.02879t}$ grams, where t is the time in years that the isotope decays. The initial amount present is 800 grams. Complete parts (a) through (c).

(a) How many grams remain after 5 years?

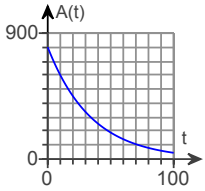
g (Type an integer or decimal rounded to two decimal places as needed.)

(b) Graph this function for $0 \leq t \leq 100$. Choose the correct graph below.

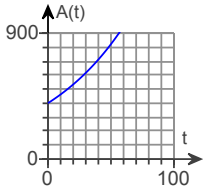
☐ A.



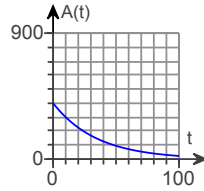
☐ B.



☐ C.



☐ D.



(c) If the half-life is the time it takes for half of the initial amount to decay, use graphical methods to estimate the half-life of this isotope. Choose the correct answer below.

☐ A. about 48.16 years

☐ B. about 9.08 years

☐ C. about 44.08 years

☐ D. about 24.08 years

4. During a 10-year period of constant inflation, the value of a \$200,000 property will increase according to the equation $v = 200,000e^{0.04t}$.
- What will be the value of the property in 6 years?
 - Use a table or graph to estimate when this property will be worth double its original value.

a. The value of the property in 6 years will be \$.
(Round to the nearest cent as needed.)

b. This property will be worth double its original value in years.
(Round to the nearest year as needed.)

5. An antique table increases in value according to the function $v(x) = 650(1.07)^x$ dollars, where x is the number of years after 1970.

- How much was the table worth in 1970?
- If the pattern indicated by the function remains valid, what was the value of the table in 1990?
- Use a table or a graph to estimate the year when this table will reach double its 1970 value.

a. The table was worth \$ in 1970.
(Round to the nearest cent as needed.)

b. The value of the table was \$ in 1990.
(Round to the nearest cent as needed.)

c. By the model, the value of this table reaches double its 1970 value in the year .

- *6. Suppose the quantity (in grams) of a radioactive substance present at time t is $Q(t) = 5000(5^{-0.7t})$, where t is measured in months.

- How much will be present in 5 months?
- How long will it take to reduce the substance to 8 grams?

a. There will be grams of the substance present in 5 months.
(Round to the nearest gram as needed.)

b. It will take months to reduce the substance to 8 grams.
(Round to the nearest month as needed.)